

EN 511: Computational Methods in Material Science

Course Code	EN 511
Course Name	Computational Methods in Material Science
L-T-P-C	1-0-6-4
Prerequisites	Instructor's consent
Intended for	UG/PG (Compulsory for MTech. in Materials and Energy Engineering, and Elective for others)
Mutual Exclusion	None .
Approval	9th Senate (May 2016)
Revised	44th BoA (Nov 2021) - revised

Course Contents

Theory

- Density functional theory, Pseudo potentials, Plane wave and Projector augmented wave methods, Exchange-correlation functionals, Self-consistent solutions, Density of states, Band structures, Optical properties, Electrical & thermal conductivities, Seebeck coefficient, Polarization, Piezoelectric tensor, Specific heat, Entropy, Free energy, Elastic tensors, Moduli of elasticity, phonon dispersion, and Machine learning. [14 Hours]

Laboratory/practical/tutorial Modules: [84 Hours]

- Part-1: Introduction to the various features of ABINIT code. Calculations of properties related to
 - Photovoltaic materials: Density of states, Band structures, & Optical properties;
 - Thermoelectric materials: Electrical conductivity, thermal conductivity, Seebeck coefficient, Specific heat, Entropy, Free energy; and
 - Piezoelectric materials: Polarization, Piezoelectric tensor, Elastic tensors & Moduli of elasticity. [48 Hours]
- Part-2: Introduction to the various features of AFLOW and JARVIS codes. Search of new materials with better
 - Electronic,
 - Optical,
 - Thermoelectric, and
 - Piezoelectric properties. [20 Hours]
- Project:** Proposing and demonstrating various scenarios for improving the properties of the existing state-of-the-art Photovoltaic, Thermoelectric, and Piezoelectric materials. [16 Hours]

Text Books

- Richard M. Martin, **Electronic Structure: Basic Theory and Practical Method**, Volume 2, Cambridge University Press, 2020.
- Tim Mueller, Aaron Gilad Kusne and Rampi Ramprasad, **Machine Learning in Materials Science: Recent Progress and Emerging Applications**, A Chapter in Reviews in Computational Chemistry, Volume 29, Editors: Abby L. Parrill and Kenny B. Lipkowitz, John Wiley & Sons, Inc., 2016

References

- Burke, Kieron, **The abc of dft**, Department of Chemistry, University of California 40 (2007).